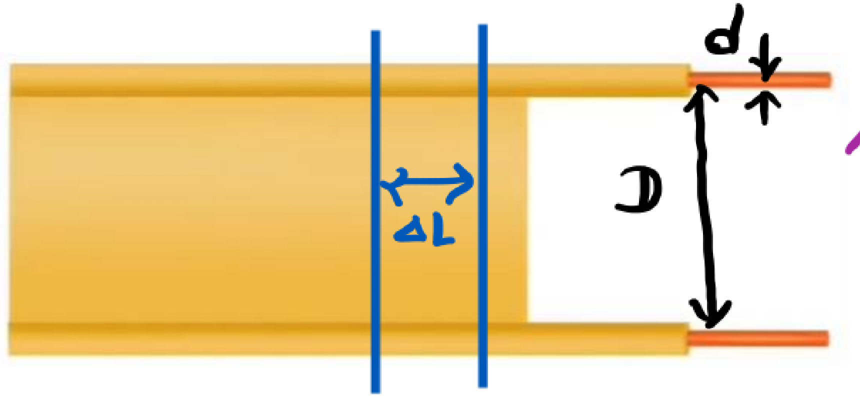
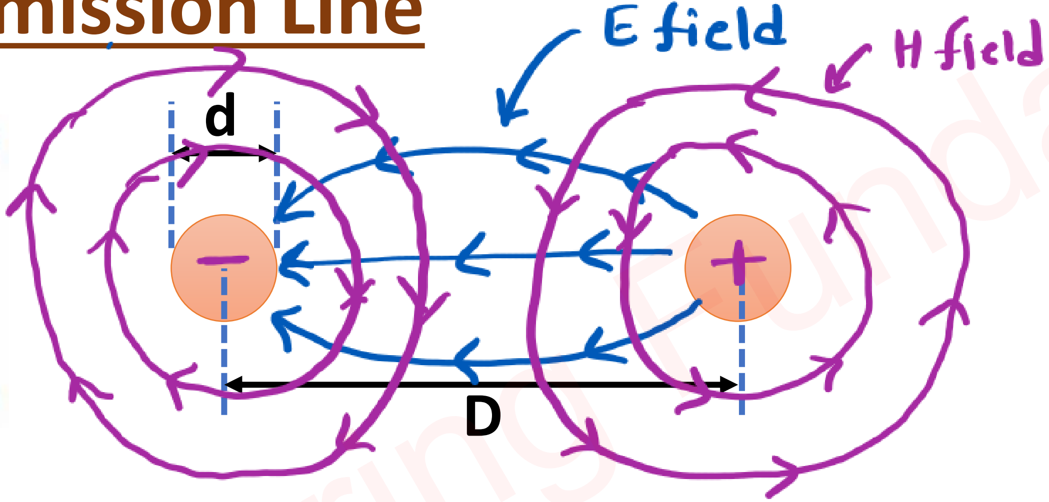
Transmission Lines of Microwave

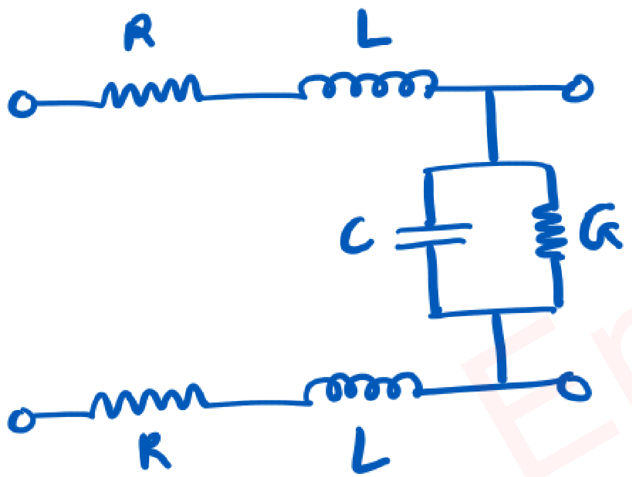
Two Parallel Wire Transmission Line





$$L = \frac{\mu}{\pi} \ln\left[\frac{2D}{d}\right]$$

$$C = \frac{\pi \epsilon}{\ln\left[\frac{2D}{d}\right]}$$



→ Characteristics Impedance

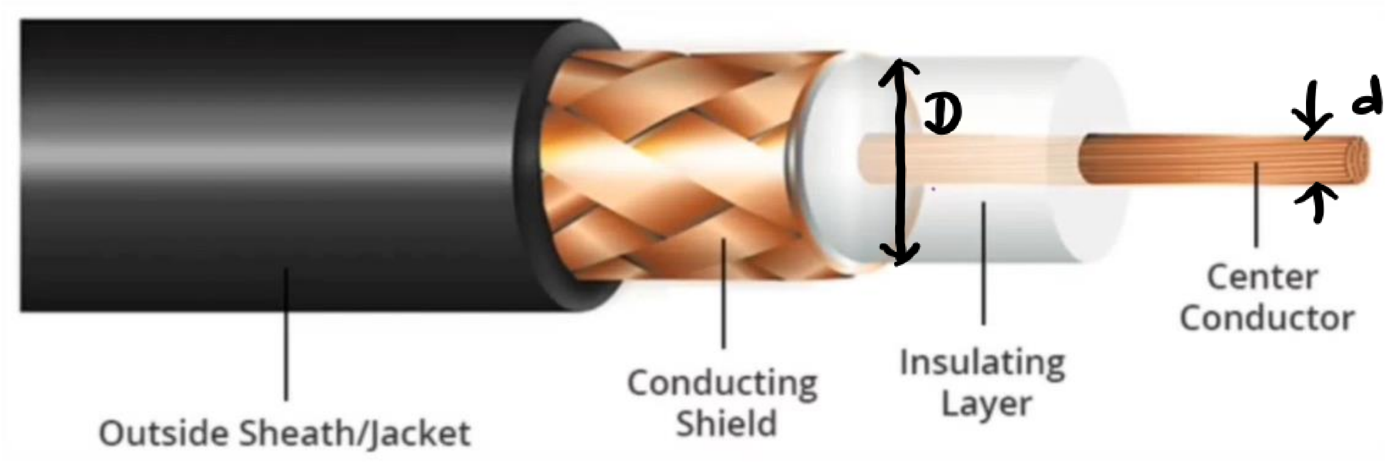
$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

→ For Lossless Tx line

$$R = G = 0$$

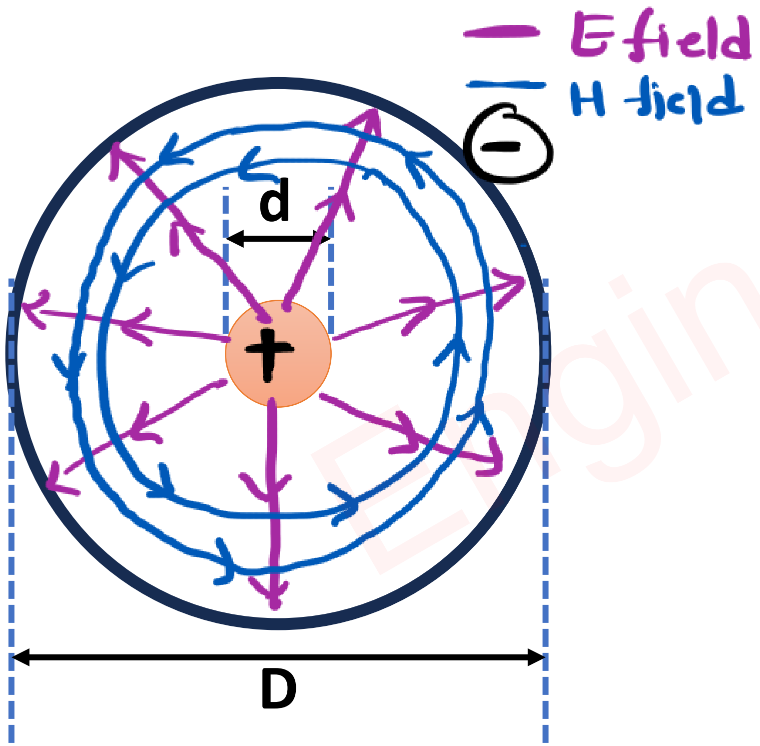
$$Z_0 = \sqrt{L/C}$$

Coaxial Transmission Line



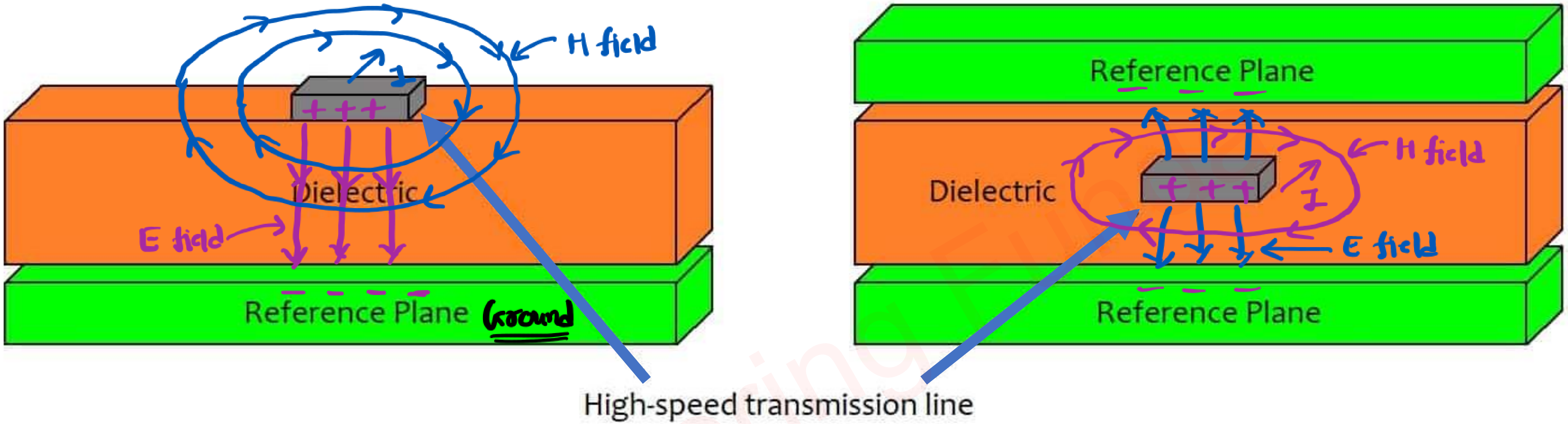
$$L = \frac{\mu}{2\pi} \ln\left[\frac{D}{d}\right]$$

$$C = \frac{2\pi\varepsilon}{\ln\left[\frac{D}{d}\right]}$$



$$\rightarrow Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

Microstrip Transmission Line

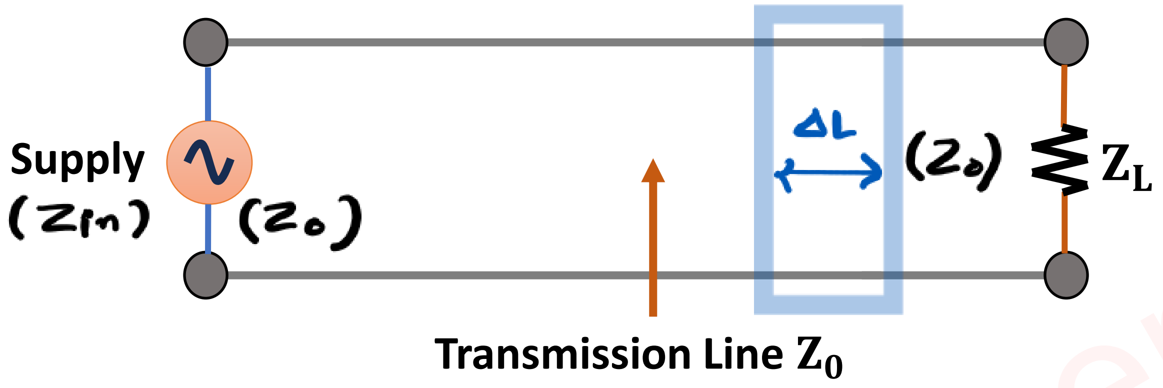


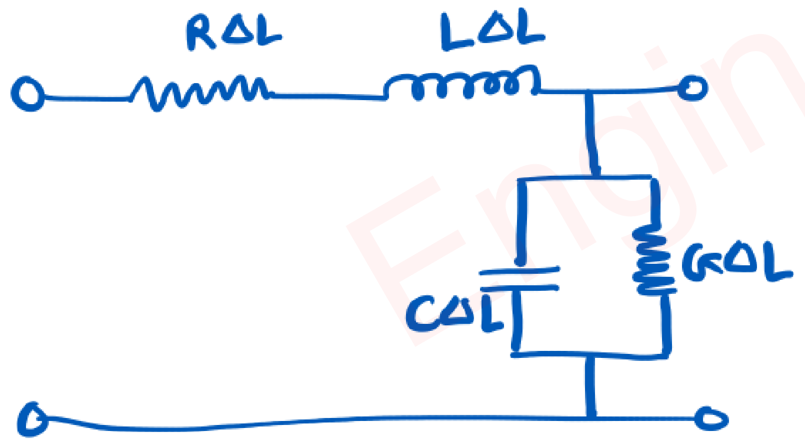
[Microstrip Transmission Line]

[Stripline Transmission Line]

Equivalent Circuit of Transmission Line

- Two Parallel Metal is Transmission Line.





↝ R & G are Lossy Components of Tx line.

Characteristic Impedance of Transmission Line

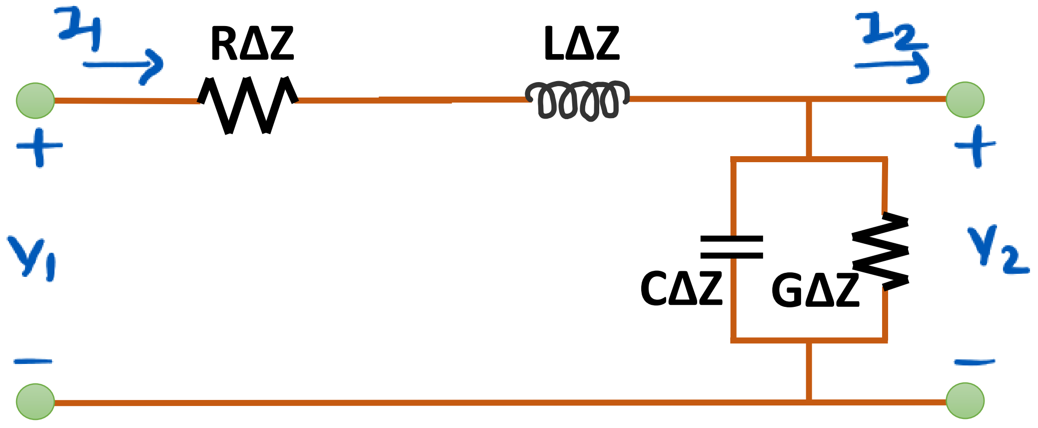
- Characteristic Impedance is the Impedance of the Transmission Line measured at I/P or O/P, Provided the Length of the Transmission Line is infinite.

$$\rightarrow Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

⤳ For Lossless Tx line. [$R = G = 0$]

$$\rightarrow Z_0 = \sqrt{L/C}$$

Transmission Lines Equations



- As per KVL, Voltage drop Across Tx Line

$$\Rightarrow \Delta V = V_2 - V_1 = -\left[R\Delta z I_1 + L\Delta z \frac{dI_1}{dt} \right]$$

$$\Rightarrow \frac{\Delta V}{\Delta z} = -\left[R + L\frac{d}{dt} \right] I_1 \quad \text{— (1)}$$

- As per KCL, Current drop Across Tx Line

$$\Rightarrow \Delta I = I_2 - I_1 = -\left[G\Delta z V_2 + C\Delta z \frac{dV_2}{dt} \right]$$

$$\Rightarrow \frac{\Delta I}{\Delta z} = -\left[G + C\frac{d}{dt} \right] V_2 \quad \text{— (2)}$$

→ Write eq.^{n} ① & ② in form of V & I.

$$\Rightarrow \frac{dV}{dz} = -\left[R + L\frac{d}{dt}\right] I \quad \text{— (3)}$$

$$\Rightarrow \frac{dI}{dz} = -\left[G + C\frac{d}{dt}\right] V \quad \text{— (4)}$$

→ Here, $\frac{d}{dt} = j\omega$

$$\Rightarrow \frac{dV}{dz} = -\left[R + j\omega L\right] I \quad \text{— (5)}$$

$$\Rightarrow \frac{dI}{dz} = -\left[G + j\omega C\right] V \quad \text{— (6)}$$

→ To get Tx line eq.^{n} differenciate eq.^{n} ⑤ w.r.t. z

$$\Rightarrow \frac{d^2V}{dz^2} = -\left[R + j\omega L\right] \frac{dI}{dz}$$

$$\Rightarrow \frac{d^2V}{dz^2} = \left[R + j\omega L\right]\left[G + j\omega C\right] V \quad \text{— (A)}$$

→ eq.^{n} (4) Tx line eq.^{n} and it is similar to wave eq.^{n}

$$\Rightarrow \frac{d^2V}{dz^2} = \gamma^2 V \quad \text{—— (7)}$$

→ γ = propagation constant

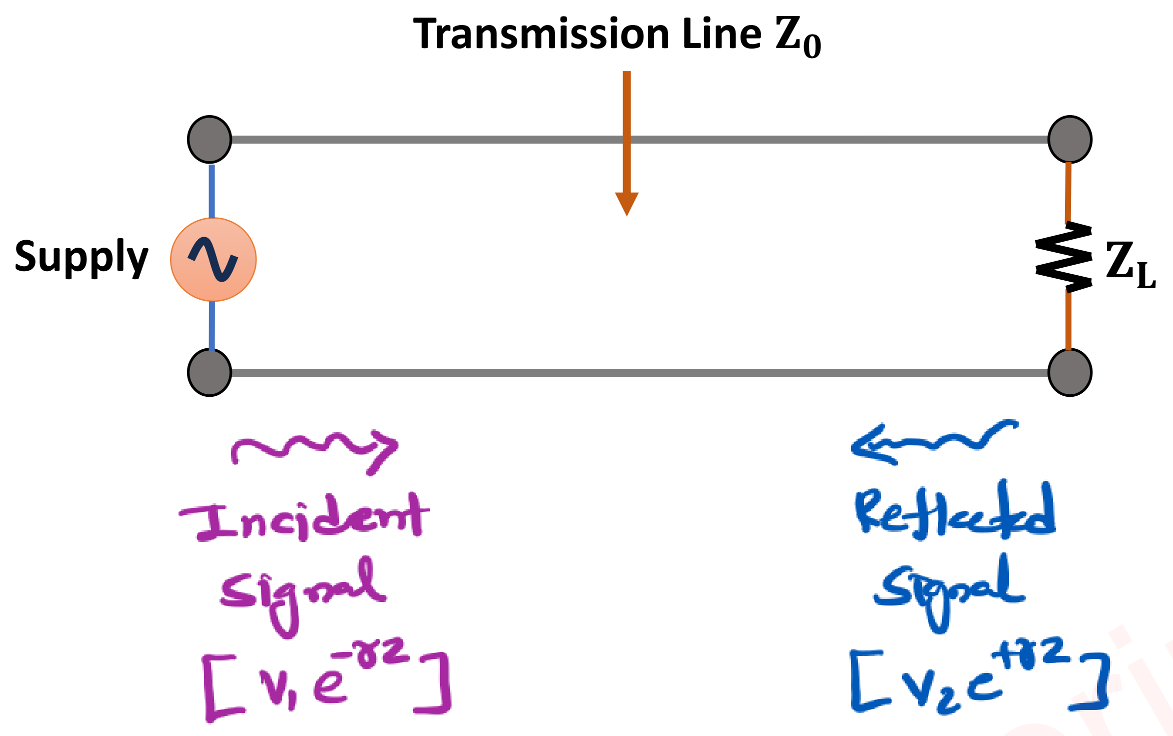
$$= \sqrt{(R + j\omega L)(G + j\omega C)}$$

→ Sol.^{n} of eq.^{n} (7) is

$$\Rightarrow V = \underbrace{V_1 e^{-\gamma z}}_{\text{Incident signal}} + \underbrace{V_2 e^{+\gamma z}}_{\text{Reflected signal.}}$$

→ Similarly,

$$\Rightarrow I = \underbrace{I_1 e^{-\gamma z}}_{\text{Incident signal}} + \underbrace{I_2 e^{+\gamma z}}_{\text{Reflected signal}}$$


Transmission Line Z_0
Supply
Z_L
Incident Signal
$[V_1 e^{-\gamma z}]$
Reflected Signal
$[V_2 e^{+\gamma z}]$

Propagation Constant, Attenuation Constant, & Phase Constant of the Transmission Line

→ From Tx line wave eq^n

$$\Rightarrow \gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$$

$$= \underline{\alpha} + j\underline{\beta}$$

Attenuation Constant

phase constant.

→ For lossless Tx line, [R = G = 0]

$$\Rightarrow \gamma = \sqrt{(0+j\omega L)(0+j\omega C)}$$

$$\Rightarrow \gamma = j\omega\sqrt{LC}$$

→ $\alpha = 0$, $\beta = \omega\sqrt{LC}$

→ α and β

$$\Rightarrow \alpha = \frac{1}{2}\left[G\sqrt{\frac{L}{C}} + R\sqrt{\frac{C}{L}}\right]$$

$$\Rightarrow \beta = \omega\sqrt{LC}\left[1 + \frac{1}{8}\left(\frac{R}{\omega L} - \frac{G}{\omega C}\right)^2\right]$$

Characteristics Impedance

- Characteristic Impedance Z_0 is the Impedance of the Transmission Line measured at I/P or O/P, Provided the Length of the Transmission Line is infinite.

⇝ Sol.^{n} of wave eq.^{n}

$$\Rightarrow V = V_1 e^{-\gamma z} + V_2 e^{+\gamma z}$$

⇝ T_x line eq.^{n}

$$\Rightarrow \frac{dV}{dz} = -(R+j\omega L) I$$

$$\Rightarrow I = \frac{-1}{R+j\omega L}\left(\frac{dV}{dz}\right)$$

$$\Rightarrow I = \frac{-1}{R+j\omega L} \frac{d}{dz}(V_1 e^{-\gamma z} + V_2 e^{\gamma z})$$

$$\Rightarrow I = \frac{-1}{R+j\omega L} (V_1(-\gamma)e^{-\gamma z} + V_2(\gamma)e^{\gamma z})$$

$$\Rightarrow I = \frac{\gamma}{R+j\omega L}\left(V_1 e^{-\gamma z} - V_2 e^{\gamma z}\right)$$

$\rightsquigarrow$ [Here, $\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$]

$$\Rightarrow I = \frac{\sqrt{(R+j\omega L)(G+j\omega C)}}{R+j\omega L}\left(V_1 e^{-\gamma z} - V_2 e^{\gamma z}\right)$$

$$\Rightarrow I = \sqrt{\frac{G+j\omega C}{R+j\omega L}}\left(V_1 e^{-\gamma z} - V_2 e^{\gamma z}\right)$$

$$\Rightarrow I = \frac{1}{Z_0}\left(V_1 e^{-\gamma z} - V_2 e^{\gamma z}\right)$$

$\rightsquigarrow$ Hence, characteritics Impedance Z_0 is

$$\Rightarrow Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

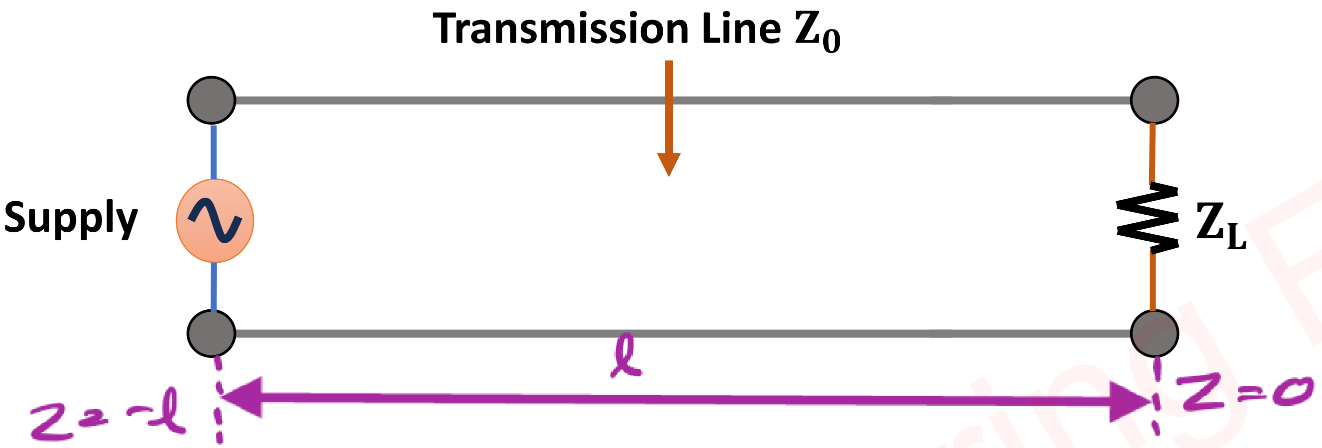
$\rightsquigarrow$ For lossless Tx line,

[$R = G = 0$]

$$\Rightarrow Z_0 = \sqrt{L/C}$$

Reflection Coefficient

- Reflection Coefficient explains the amount of reflection of the signal from the load.



→ Based on Voltage & Current eqⁿ of Tx line

$$\Rightarrow V = \underbrace{V_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{V_2 e^{\gamma z}}_{\text{Reflected Signal}} \qquad \Rightarrow I = \underbrace{I_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{I_2 e^{\gamma z}}_{\text{Reflected Signal}}$$

→ So, above eqⁿ are :

$$\Rightarrow V = V_i e^{-\gamma z} + V_r e^{\gamma z} \qquad \Rightarrow I = I_i e^{-\gamma z} + I_r e^{\gamma z}$$

- At load $z = 0$

$$\Rightarrow V_L = V_i + V_r \qquad \Rightarrow I_L = I_i + I_r$$

- At Source $z = -l$

$$\Rightarrow V_S = V_i e^{\gamma l} + V_r e^{-\gamma l} \qquad \Rightarrow I_S = I_i e^{\gamma l} + I_r e^{-\gamma l}$$

- Characteristics Impedance Z_0

$$\Rightarrow Z_0 = \frac{V_i}{I_i} = -\frac{V_r}{I_r}$$

- Reflection Coefficient γ

$$\Rightarrow \gamma = \frac{V_r}{V_i} = -\frac{I_r}{I_i}$$

- Load Impedance Z_L

$$\Rightarrow Z_L = \frac{V_L}{I_L} = \frac{V_i + V_r}{I_i + I_r}$$

$$\Rightarrow Z_L = \frac{V_i}{I_i}\left(\frac{1+(V_r/V_i)}{1+(I_r/I_i)}\right)$$

$$\Rightarrow Z_L = Z_0\left(\frac{1+\gamma}{1-\gamma}\right)$$

$$\Rightarrow Z_L(1-\gamma) = Z_0(1+\gamma)$$

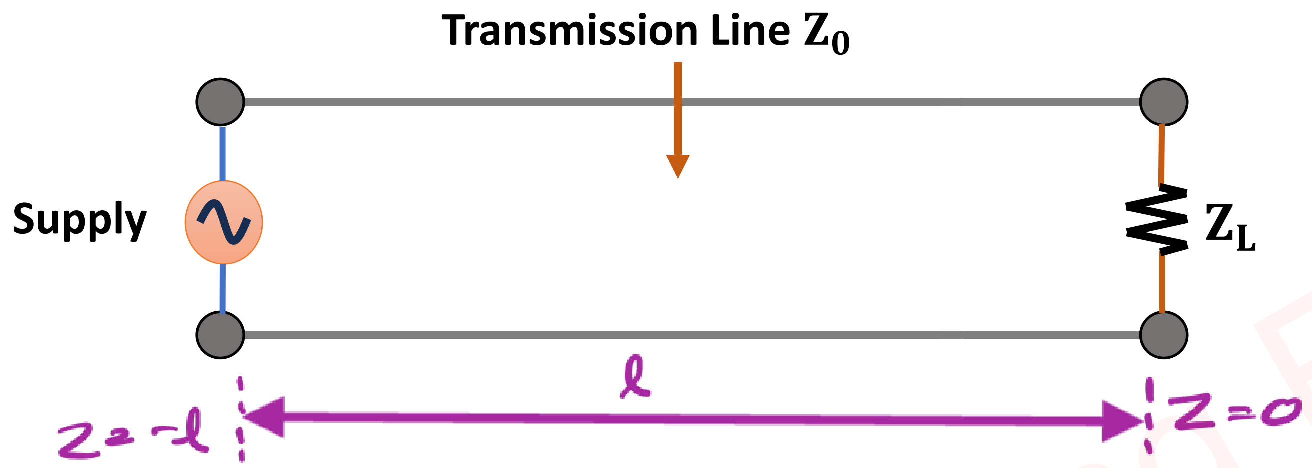
$$\Rightarrow Z_L - Z_0 = \gamma(Z_L + Z_0)$$

$$\Rightarrow \boxed{\gamma = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

~ For perfect Impedance matching $Z_L = Z_0$

$$\Rightarrow \gamma = \frac{0}{2Z_L} = 0$$

Input Impedance



- Based on Voltage & Current eqⁿ of Tx line

$\Rightarrow V = \underbrace{V_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{V_2 e^{\gamma z}}_{\text{Reflected Signal}}$ $\qquad \Rightarrow I = \underbrace{I_1 e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{I_2 e^{\gamma z}}_{\text{Reflected Signal}}$

- So, above eqⁿ are :

$\Rightarrow V = V_i e^{-\gamma z} + V_r e^{\gamma z}$ $\qquad \Rightarrow I = I_i e^{-\gamma z} + I_r e^{\gamma z}$

- At load $z = 0$

$\Rightarrow V_L = V_i + V_r$ $\qquad \Rightarrow I_L = I_i + I_r$

~ At Source $z = -l$

$$\Rightarrow V_s = V_i e^{\gamma l} + V_r e^{-\gamma l} \qquad \Rightarrow I_s = I_i e^{\gamma l} + I_r e^{-\gamma l}$$

~ Characteristics Impedance Z_0

$$\Rightarrow Z_0 = \frac{V_i}{I_i} = -\frac{V_r}{I_r}$$

~ Reflection Coefficient γ

$$\Rightarrow \gamma = \frac{V_r}{V_i} = -\frac{I_r}{I_i}$$

~ Input Impedance Z_{in}

$$\Rightarrow Z_{in} = \frac{V_s}{I_s} = \frac{V_i e^{\gamma l} + V_r e^{-\gamma l}}{I_i e^{\gamma l} + I_r e^{-\gamma l}} = \frac{V_i \left(e^{\gamma l} + (V_r/V_i) e^{-\gamma l} \right)}{I_i \left(e^{\gamma l} + (I_r/I_i) e^{-\gamma l} \right)}$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{e^{\gamma l} + \gamma e^{-\gamma l}}{e^{\gamma l} - \gamma e^{-\gamma l}} \right]$$

[Reflection Coefficient $\gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0}$]

$$\Rightarrow Z_{in} = Z_0 \left[\frac{e^{\gamma l} + \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-\gamma l}}{e^{\gamma l} - \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-\gamma l}} \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{e^{\gamma l}(Z_L + Z_0) + (Z_L - Z_0) e^{-\gamma l}}{e^{\gamma l}(Z_L + Z_0) - (Z_L - Z_0) e^{-\gamma l}} \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L(e^{\gamma l} + e^{-\gamma l}) + Z_0(e^{\gamma l} - e^{-\gamma l})}{Z_L(e^{\gamma l} - e^{-\gamma l}) + Z_0(e^{\gamma l} + e^{-\gamma l})} \right]$$

[For lossless Tx line $\alpha = 0$, $\gamma = \alpha + j\beta = j\beta$]

$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L(e^{j\beta l} + e^{-j\beta l}) + Z_0(e^{j\beta l} - e^{-j\beta l})}{Z_L(e^{j\beta l} - e^{-j\beta l}) + Z_0(e^{j\beta l} + e^{-j\beta l})} \right]$$

$$\left[\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j} \; , \quad \cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2} \right]$$

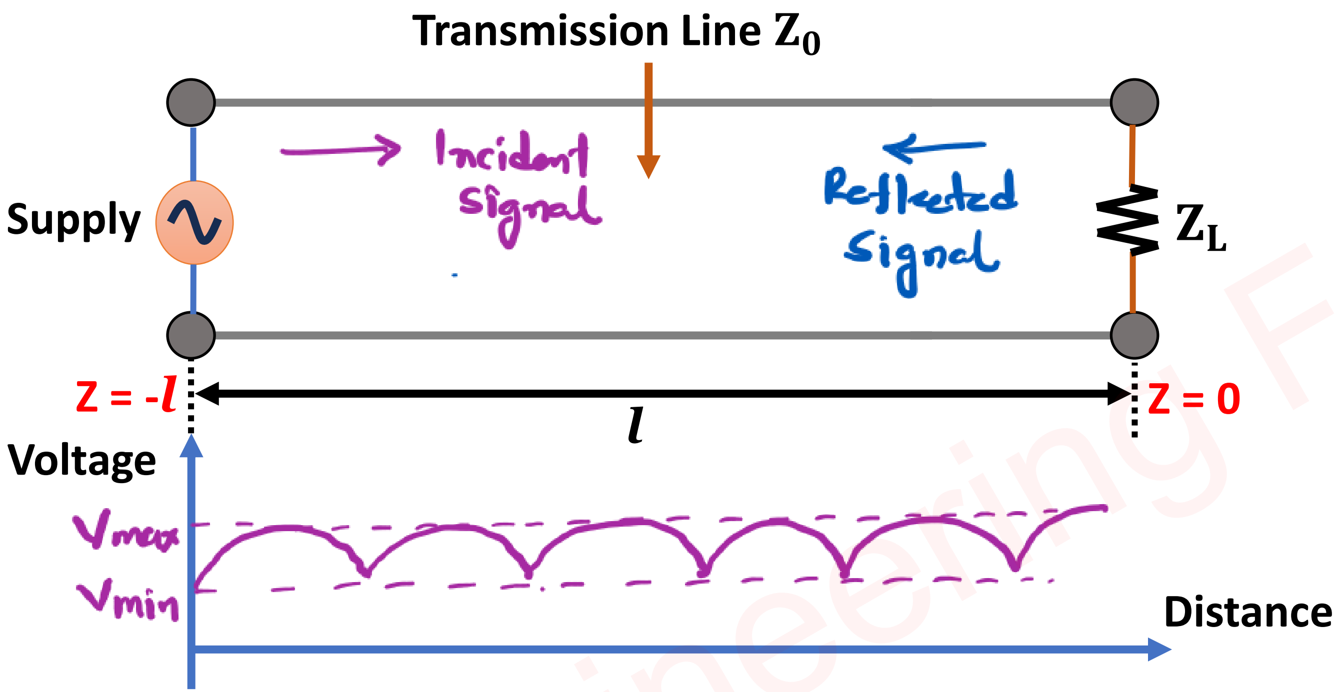
$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L (\cancel{2} \cos\beta l) + Z_0 (\cancel{2} j \sin\beta l)}{Z_L (\cancel{2} j \sin\beta l) + Z_0 (\cancel{2} \cos\beta l)} \right]$$

$$\boxed{\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L + j Z_0 \tan\beta l}{Z_0 + j Z_L \tan\beta l} \right]}$$

[Normalized Impedance, $z_{in} = \frac{Z_{in}}{Z_0}$, $z_L = \frac{Z_L}{Z_0}$]

$$\boxed{\Rightarrow z_{in} = \frac{z_L + j \tan\beta l}{1 + j z_L \tan\beta l}}$$

VSWR, Z_{MAX} and Z_{MIN} of Tx Line



↝ Voltage eq.ⁿ of Tx line

$$\Rightarrow V = \underbrace{V_i e^{-\gamma z}}_{\text{Incident signal}} + \underbrace{V_R e^{+\gamma z}}_{\text{Reflected signal.}}$$

→ Max. Voltage V_{max}

$\Rightarrow V_{max} = V_i + V_R$

→ Min. Voltage V_{min}

$\Rightarrow V_{min} = V_i - V_R$

→ Voltage Standing Wave Ratio (VSWR)

$$\Rightarrow VSWR = \frac{V_{max}}{V_{min}} = \frac{V_i + V_R}{V_i - V_R}$$

$$\Rightarrow VSWR = \frac{\cancel{V_i}(1 + V_R/V_i)}{\cancel{V_i}(1 - V_R/V_i)}$$

$$\Rightarrow \boxed{VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}}$$

→ Range of Reflection Coefficient is from 0 to 1.

Range of VSWR is from 1 to ∞.

→ Reflection Coefficient, $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$

→ If $Z_L > Z_0 \Rightarrow \Gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0} \Rightarrow VSWR = \dfrac{Z_L}{Z_0}$

→ If $Z_0 > Z_L \Rightarrow \Gamma = \dfrac{Z_0 - Z_L}{Z_L + Z_0} \Rightarrow VSWR = \dfrac{Z_0}{Z_L}$

→ Max. Impedance of Tx line.

$$\Rightarrow Z_{max} = \frac{V_{max}}{I_{min}}$$

$$= \frac{V_{max}}{V_{min}} \times \frac{V_{min}}{I_{min}}$$

$$= VSWR \times Z_0$$

→ Min. Impedance of Tx line

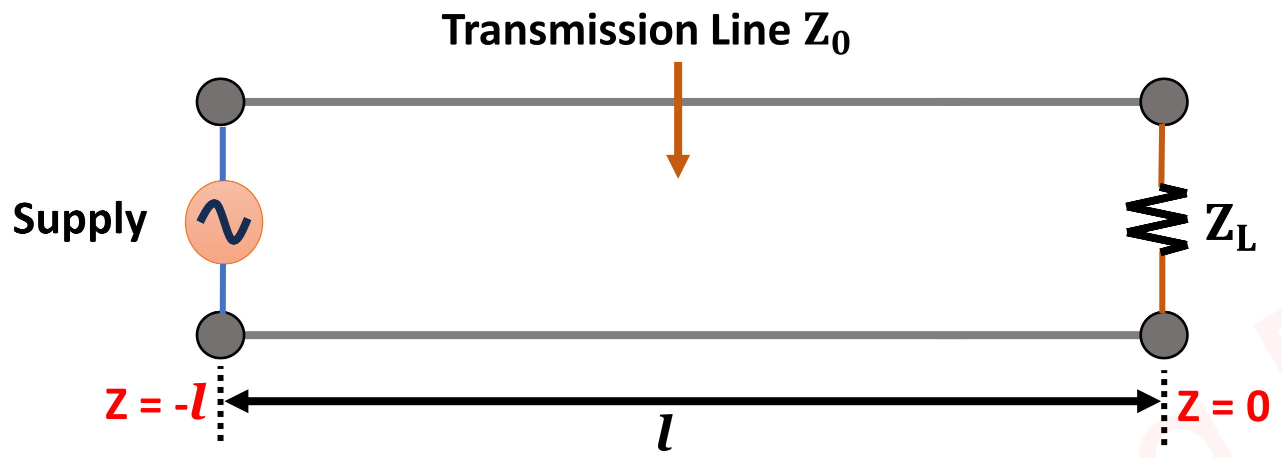
$$\Rightarrow Z_{min} = \frac{V_{min}}{I_{max}}$$

$$= \frac{V_{min}}{V_{max}} \; \frac{V_{max}}{I_{max}}$$

$$= \frac{1}{VSWR} \times Z_0$$

$$= \frac{Z_0}{VSWR}$$

Transmittance Coefficient



- Voltage eq.^{n} of Tx line

$$\Rightarrow V = \underbrace{V_i e^{-\gamma z}}_{\text{Incident Signal}} + \underbrace{V_r e^{\gamma z}}_{\text{Reflected Signal}}$$

- At load, $z = 0$

$$\Rightarrow V_L = V_i + V_r$$

~ Transmitted signal

$$\Rightarrow v_t = V_i + V_r$$

~ Incident signal

$$v_i = v_i$$

~ Transmittance Coefficient $\tau = \dfrac{\text{Transmitted signal}}{\text{Incident signal}}$

$$= \frac{v_t}{v_i} = \frac{V_i + V_r}{V_i}$$

$$= 1 + \left(\frac{V_r}{V_i}\right)$$

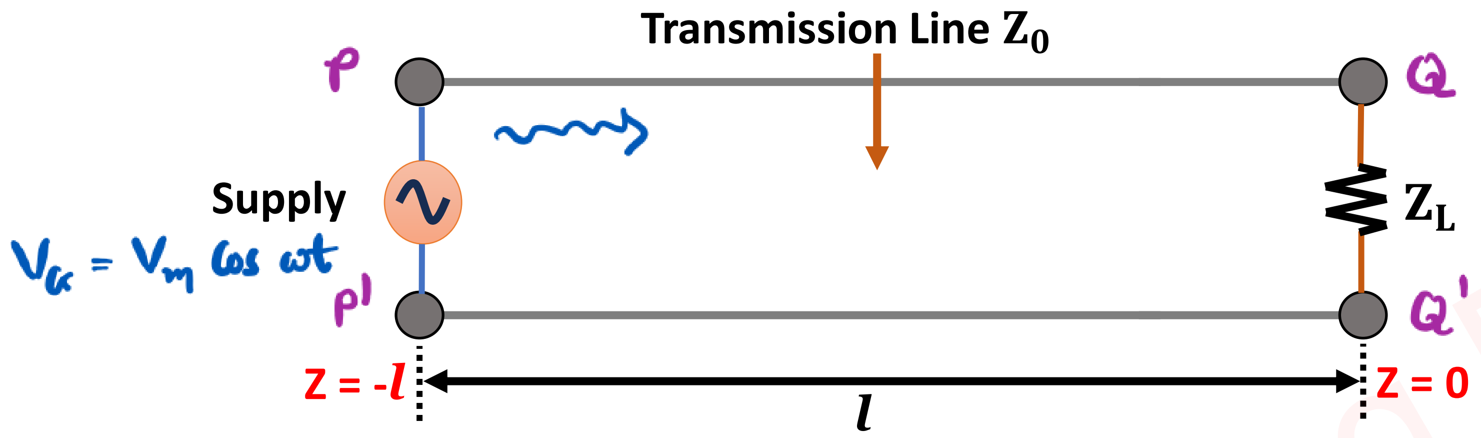
$$\boxed{\tau = 1 + \gamma}$$

~ [Reflection Coefficient $\gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0}$]

$$\Rightarrow \tau = 1 + \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L + \cancel{Z_0} + Z_L - \cancel{Z_0}}{Z_L + Z_0}$$

$$\boxed{\Rightarrow \tau = \frac{2Z_L}{Z_L + Z_0}}$$

Generation of Phase Delay



~ Let's assume 100 % Impedance Matching of Supply & Tx line.

$\Rightarrow V_{PP'} = V_m \cos \omega t$

~ Let's assume 100 % Impedance matching of Tx line & Load.

$\Rightarrow V_{QQ'} = V_m \cos \omega (t + \Delta t)$

$\left[v_p = \frac{L}{\Delta t} \Rightarrow \Delta t = \frac{L}{v_p} \right]$

$\Rightarrow V_{QQ'} = V_m \cos \left(\omega t + \frac{\omega L}{v_p} \right)$

$$\Rightarrow V_{QQ'} = V_m \cos(\omega t + \phi)$$

→ Here, phase $\phi = \dfrac{\omega L}{v_p}$

→ $f = 1\ \text{KHz}$, $L = 1\,m$, $v_p = 2\times10^{8}\ m/sec$

$$\phi = \frac{2\pi f L}{v_p} = \frac{\cancel{2}\pi \times 10^{3} \times 1}{\cancel{2}\times 10^{8}} = 3.14\times 10^{-5}\ \text{rad}$$

→ $f = 1\ \text{KHz}$, $L = 1\,m$, $v_p = 2\times10^{8}\ m/sec$

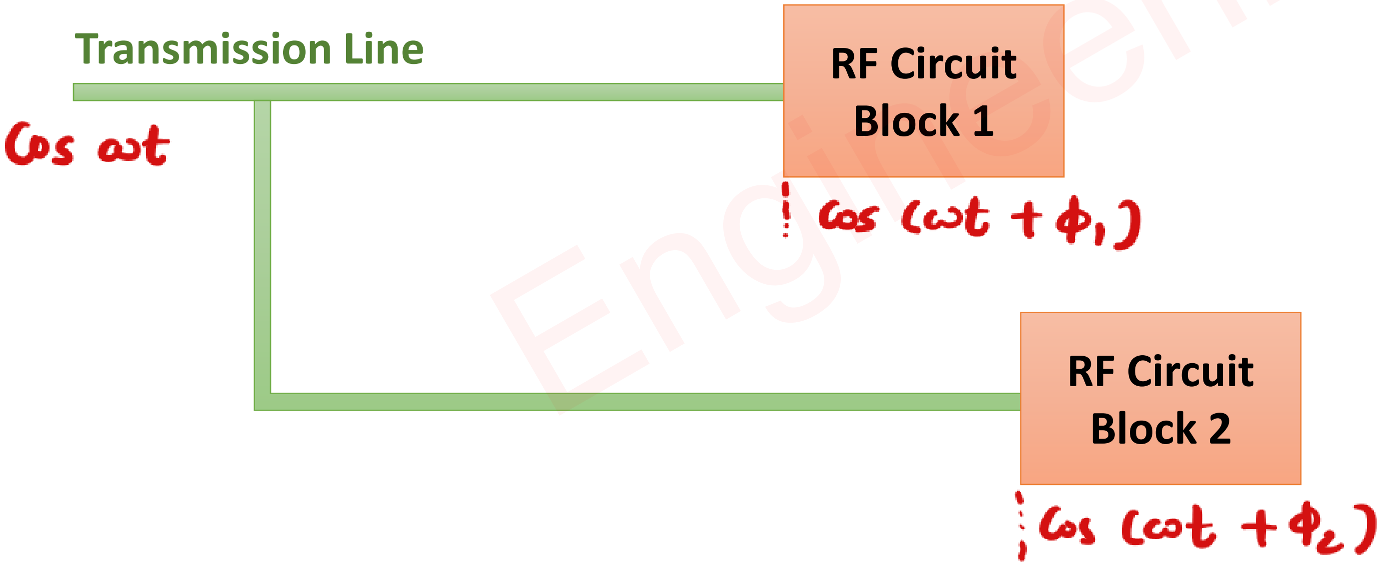
$$\phi = \frac{2\pi f L}{v_p} = \frac{\cancel{2}\pi \times 10^{9} \times 1}{\cancel{2}\times 10^{8}} = 31.4\ \text{rad}$$

Microwave Effects on Tx Line

❑ Phase Delay

- Phase Delay happens due to the length of the Transmission Line.
- Phase Delay generates synchronization issues in the RF Circuits.
- Phase Delay also depends on the Frequency of Microwave Signals.

$$\text{Phase } \phi = \frac{2\pi f L}{v_p}$$



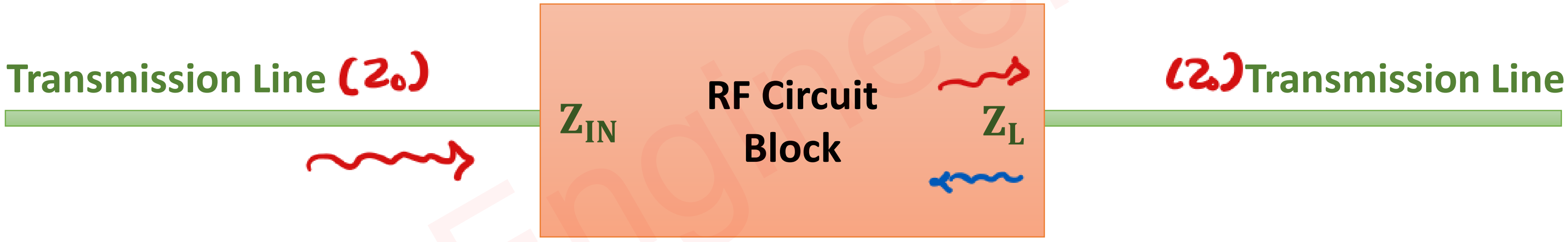
❑ Reflection of Signals

- Reflection of Signals happens due to the Impedance mismatch.

$$\text{Reflection Coefficient } \Upsilon = \frac{Z_L - Z_0}{Z_L + Z_0}$$

- Characteristics Impedance Z_0 depends on the operating frequency.

$$\text{Characteristics Impedance } Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

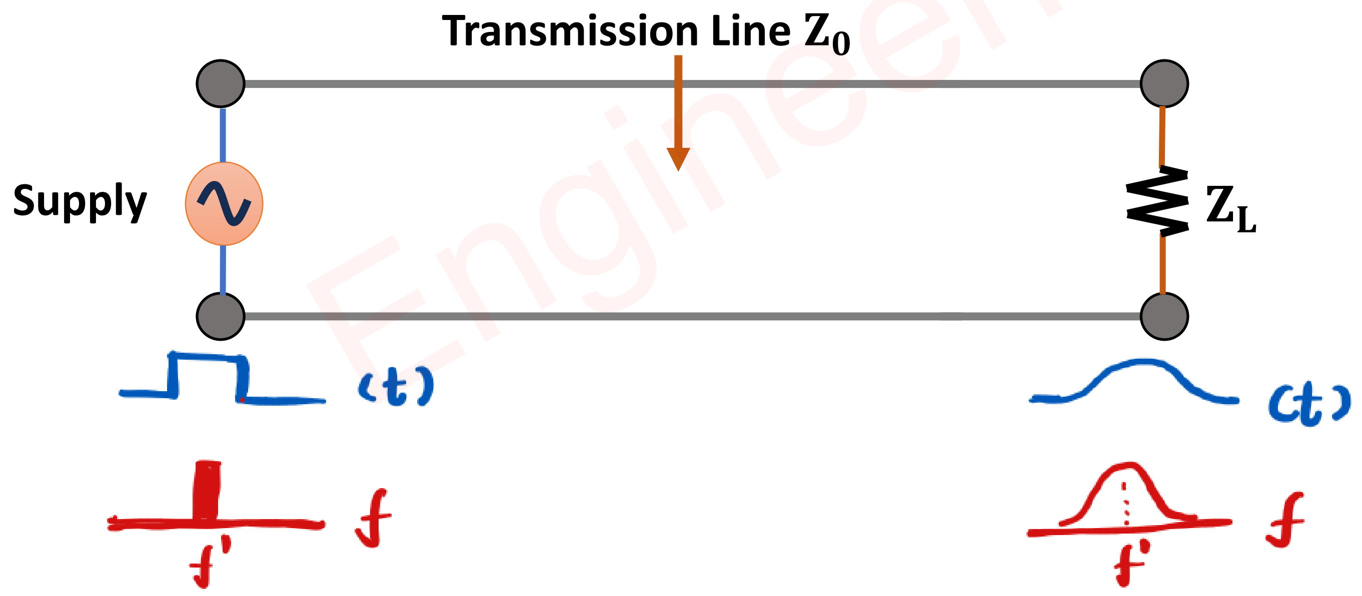


❑ Power Loss

- Power Loss in Transmission Lines happens due to Conducting and Insulating Materials.
- In Insulating Materials, Microwave signal attenuation happens.
- In Conducting Materials, Microwave signals have issues due to skin effects and signal absorption.
- Power Loss increases with an increase in the frequency of Microwave Signal.

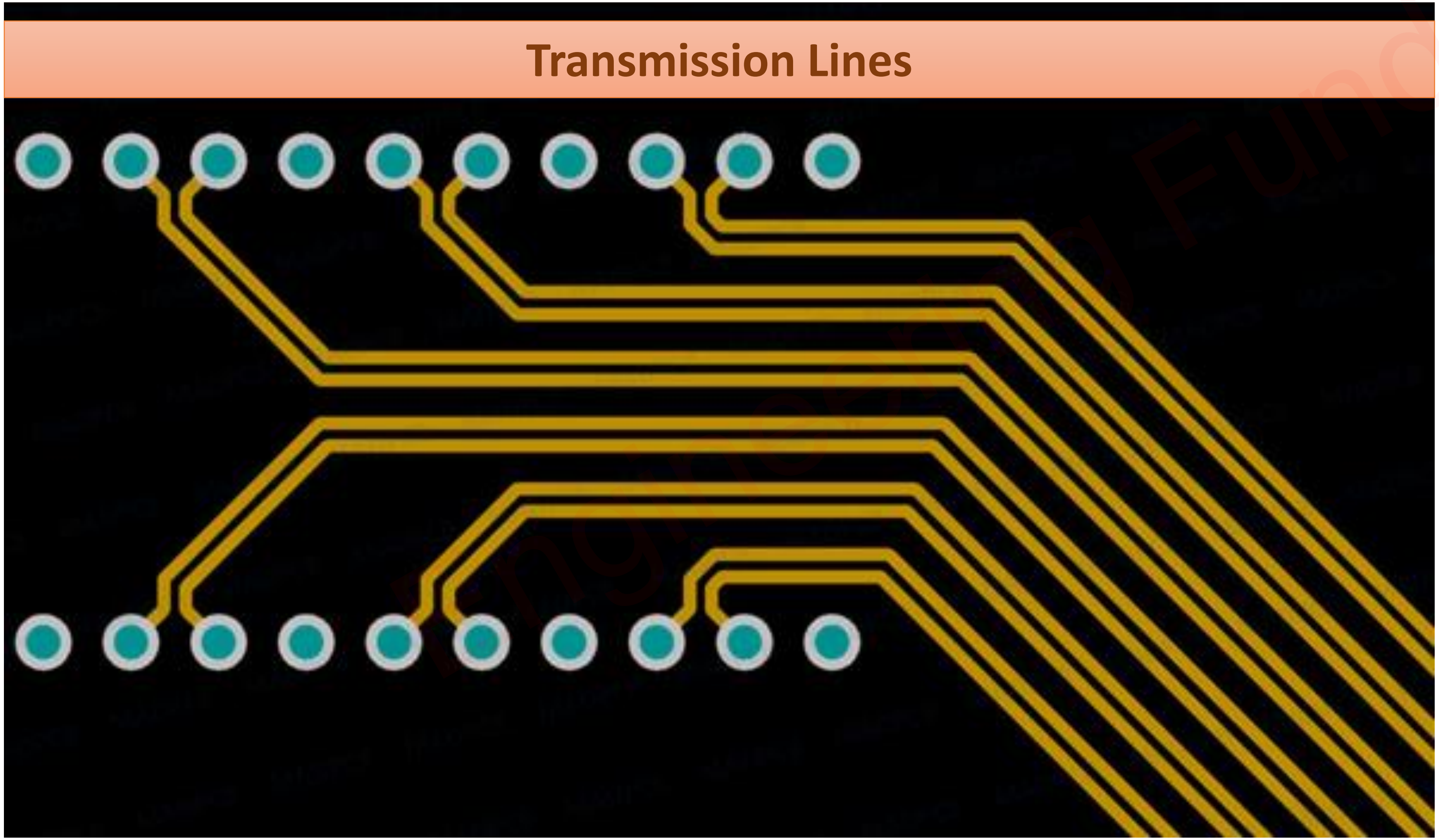
❑ Signal Dispersion

- Signal Dispersion results in the broadening of transmitted pulses.
- Due to the broadening of pulses, signals have more frequency components.
- Signal Dispersion also depends on the frequency and the Length of the Transmission Line.



❑ Coupling Loss

- EM signal couples in between transmission lines and RF Circuits.
- Coupling Loss also depends on the spacing between transmission lines and Isolation between Transmission Lines.



Limitations of Tx Line

❑ Separation Between Two Parallel wires.

- At Lower Frequencies, the wavelengths will be higher, Hence we need a large separation between two parallel wires.
- At Higher Frequencies, the wavelengths will be shorter, Hence we need a small separation between two parallel wires.
- So, After the fabrication of the Transmission Line, we can not change the operating frequency of the Transmission Line.

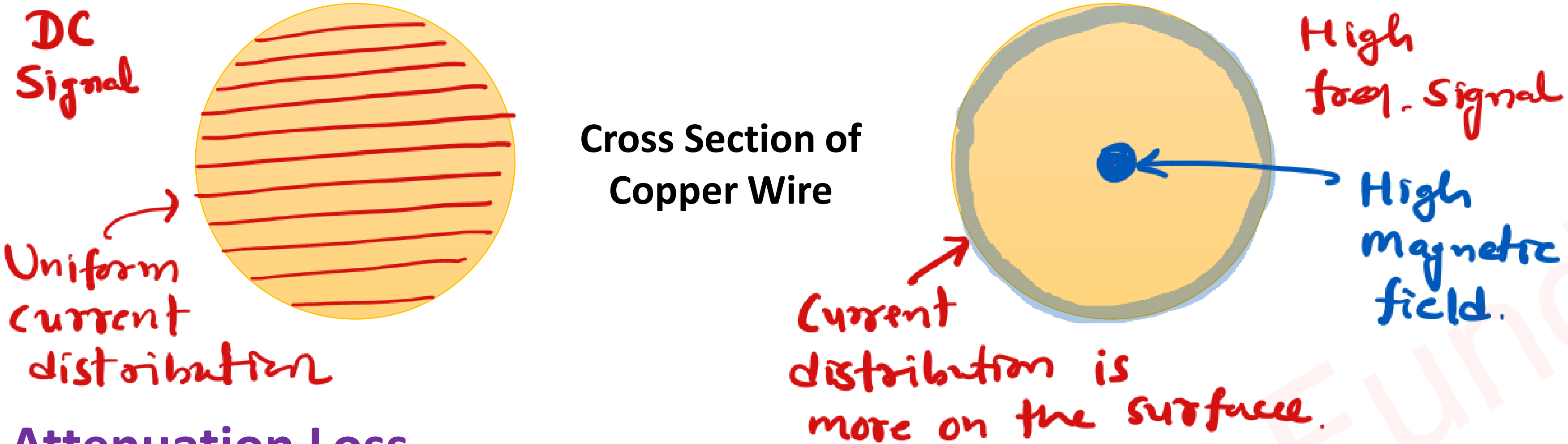
$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

❑ Copper Loss.

- Copper Loss depends on the Surface area of the Metal (Copper).
- The lower the surface area, the higher the Copper Loss with the Transmission Line.

❑ Skin Effect Loss.

- At Microwave frequencies, the skin effect creates losses of signal.
- At Microwave frequencies, Current distribution happens on the perimeter of the wire, which means the current distribution repeals towards the surface of the wire.
- If we further increase the frequency, signal is getting radiated in the free space.
- So, at higher frequencies, loss due to skin effect is higher.



❑ Attenuation Loss.

- At Microwave frequencies, the wavelength is shorter, so the separation is also short between two wires, which leads to higher attenuation of signal.
- At Microwave frequencies, due to attenuation of signal, if we increase the power, there is a possibilities of dielectric breakdown.

❑ Radiation and Induction Loss.

- At Microwave frequencies, the wavelength is shorter, so it may radiate EM waves.
- At Microwave frequencies, Lumped parameters like inductance is included, which results in induction loss.

❑ Power Handling Capacity.

- Power Handling capacity is limited due to dielectric breakdown.

Transmission Line Losses

❑ Attenuation Loss.

- It happens due to the absorption of signal in the Transmission Line.
- Major Attenuation happens in dielectric material.

$$L_{att} = 10 \log \left[\frac{E_i - E_r}{E_t} \right]$$

$$\because \; E \propto V^2 \Rightarrow E_i \propto V_i^2 , \; E_r \propto V_r^2$$

$$E_t = [\,|V_i^2| - |V_r^2|\,] e^{-2\alpha l}$$

$$\Rightarrow L_{att} = 10 \log \left[\frac{\cancel{|V_i^2| - |V_r^2|}}{[\cancel{|V_i^2| - |V_r^2|}] e^{-2\alpha l}} \right]$$

$$\Rightarrow L_{att} = 10 \log [e^{2\alpha l}]$$

$$\Rightarrow L_{att} = 20 \alpha l \log e$$

$$\Rightarrow L_{att} = 8.686 \alpha l$$

❑ Reflection Loss.

- Reflection Loss happens due to the mismatch of the Transmission Line.

$$L_{Ref} = 10 \log \left[\frac{E_i}{E_i - E_r} \right]$$

$$= 10 \log \left[\frac{|V_i^2|}{|V_i^2| - |V_r^2|} \right]$$

$$= 10 \log \left[\frac{1}{1 - \left| \left(\frac{V_r}{V_i} \right)^2 \right|} \right]$$

→ Reflection Coefficient $\Gamma = \frac{V_r}{V_i}$

$$\Rightarrow L_{Ref} = 10 \log \left[\frac{1}{1 - |\Gamma^2|} \right]$$

❑ Transmission Loss.

- Transmission Loss is a loss in the Transmission Line.

$$L_{Trans} = 10 \log \left[\frac{E_i}{E_t} \right]$$

$$= 10 \log \left[\frac{E_i}{E_i - E_r} \times \frac{E_i - E_r}{E_t} \right]$$

$$= 10 \log \left[\frac{E_i}{E_i - E_r} \right] + 10 \log \left[\frac{E_i - E_r}{E_t} \right]$$

$$= L_{Ref} + L_{Att}$$

$$L_{Trans} = 10 \log \left[\frac{1}{1 - |\Gamma^2|} \right] + 8.686\, \alpha l$$

❑ Return Loss.

- Return Loss happens at the port of the Transmission Line.
- It happens due to an Impedance mismatch at the port of the Transmission Line.

$$L_{Ret} = 10 \log \left[\frac{E_r}{E_i} \right]$$

$$= 10 \log \left[\frac{|V_r^2|}{|V_i^2|} \right]$$

⇝ Reflection Coefficient, $\Gamma = \frac{V_r}{V_i}$

$$\Rightarrow L_{Ret} = 10 \log |\Gamma^2|$$

$$\Rightarrow L_{Ret} = 20 \log |\Gamma|$$

Insertion Loss.

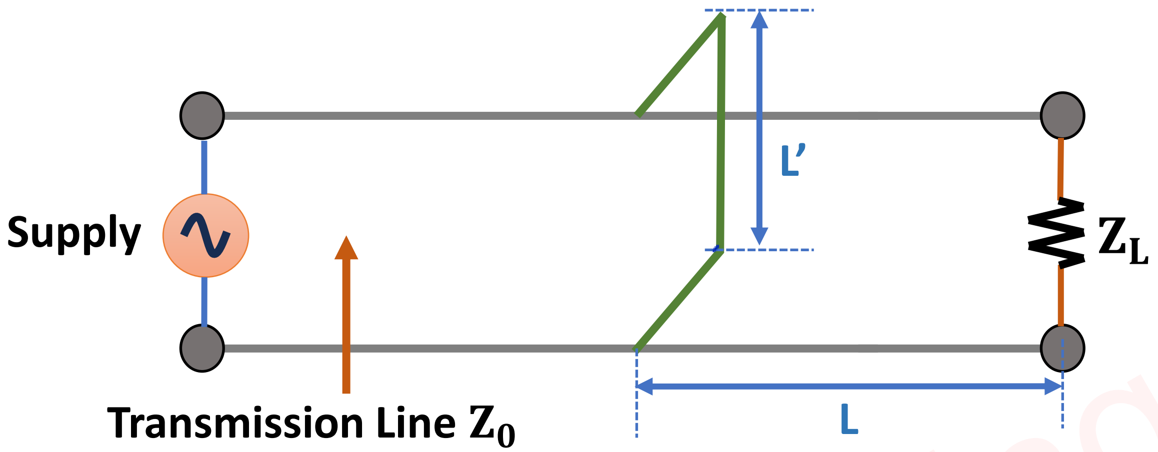
- Insertion Loss happens due to the insertion of the Transmission Line.

→ E_1 is energy with Tx line

E_2 is energy without Tx line

$$\Rightarrow L_{ins} = 10 \log \left[\frac{E_2}{E_1} \right]$$

Impedance Matching using Single Stub



❑ Basics of Impedance Matching

- For Maximum power transfer to the load Z_L, Impedance of Z_L and Z_0 should get matched, else there will be a reflection of signal.
- Reflection coefficient can be identified by:

$$\Upsilon = \frac{V_r}{V_i} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

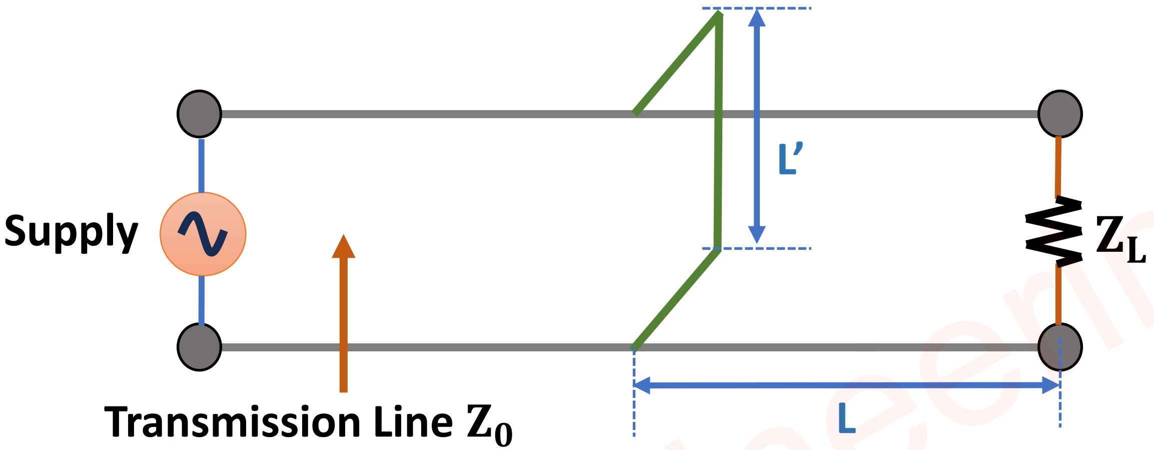
- To minimize signal reflection, we can use a single stub (Short Circuited wire), which can provide impedance matching of Transmission line with load.

Basics of Single Stub Impedance Matching

- The stub (L) location should be such that Input Admittance should be $Y_1 = 1 + jX$.
- The Length of the Stub (L') should be such that the admittance of the stub should be $Y_2 = -jX$.
- So, Total Admittance is

$$Y = Y_1 + Y_2 = 1$$

$\rightarrow Y = 1 = \frac{Y_L}{Y_0} \Rightarrow Y_L = Y_0 \Rightarrow Z_L = Z_0$



→ Input Impedance

$$\Rightarrow Z_{in} = \frac{Z_L + j\tan\beta l}{1 + jZ_L \tan\beta l}$$

$$[\; Y = \tfrac{1}{2} \;]$$

$$\Rightarrow \left(\frac{1}{Y_{in}}\right) = \frac{(1/Y_L) + j\tan\beta l}{1 + j(1/Y_L)\tan\beta l}$$

$$\Rightarrow Y_{in} = \frac{Y_L + j\tan\beta l}{1 + jY_L \tan\beta l} \times \frac{1 - jY_L \tan\beta l}{1 - jY_L \tan\beta l}$$

$$\Rightarrow Y_{in} = \frac{Y_L + Y_L \tan^2\beta l + j(\tan\beta l - Y_L^2 \tan\beta l)}{1 + Y_L^2 \tan^2\beta l}$$

$$\Rightarrow Y_{in} = \frac{Y_L(1+\tan^2\beta l)}{1+Y_L^2\tan^2\beta l} + j\,\frac{(1-Y_L^2)\tan\beta l}{1+Y_L^2\tan^2\beta l}$$

~> Here, Real component = 1

$$\Rightarrow \frac{Y_L(1+\tan^2\beta l)}{1+Y_L^2\tan^2\beta l} = 1$$

$$\Rightarrow Y_L + Y_L\tan^2\beta l = 1 + Y_L^2\tan^2\beta l$$

$$\Rightarrow Y_L - 1 = (Y_L^2 - Y_L)\tan^2\beta l$$

$$\Rightarrow \cancel{Y_L - 1} = Y_L\,\cancel{(Y_L - 1)}\,\tan^2\beta l$$

$$\Rightarrow \tan^2\beta l = 1/Y_L$$

$$\Rightarrow \tan \beta l = \frac{1}{\sqrt{Y_L}}$$

$$\Rightarrow \tan \beta l = \sqrt{Z_L}$$

$$\Rightarrow \tan \beta l = \sqrt{\frac{Z_L}{Z_0}}$$

$$\Rightarrow \beta l = \tan^{-1}\sqrt{\frac{Z_L}{Z_0}} \qquad \left[\because \beta = \frac{2\pi}{\lambda}\right]$$

$$\Rightarrow \boxed{l = \frac{\lambda}{2\pi} \tan^{-1}\sqrt{\frac{Z_L}{Z_0}}}$$

~> Here, we have short circuited stub,

$$\Rightarrow Z_{sc} = j Z_0 \tan \beta l'$$

$$\Rightarrow Z_{sc}/Z_0 = j \tan \beta l'$$

$$\Rightarrow z_{sc} = j \tan \beta l'$$

$$\Rightarrow Y_{sc} = -j \cot \beta l'$$

$$\leadsto \quad Y_{sc} = -jX, \quad \text{Here } X = \frac{(1-Y_L^2)\tan\beta l}{1+Y_L^2\tan^2\beta l}$$

$$\Rightarrow \cot\beta l' = \frac{(1-Y_L^2)\tan\beta l}{1+Y_L^2\tan^2\beta l}$$

$$\Rightarrow \cot\beta l' = \frac{(1-Y_L^2)\left(1/\sqrt{Y_L}\right)}{1+Y_L^{\cancel{2}}\left(1/\cancel{Y_L}\right)}$$

$$\Rightarrow \cot\beta l' = (1-Y_L)\left(1/\sqrt{Y_L}\right)$$

$$\Rightarrow \cot\beta l' = \left(1-\frac{Z_0}{Z_L}\right)\sqrt{\frac{Z_L}{Z_0}}$$

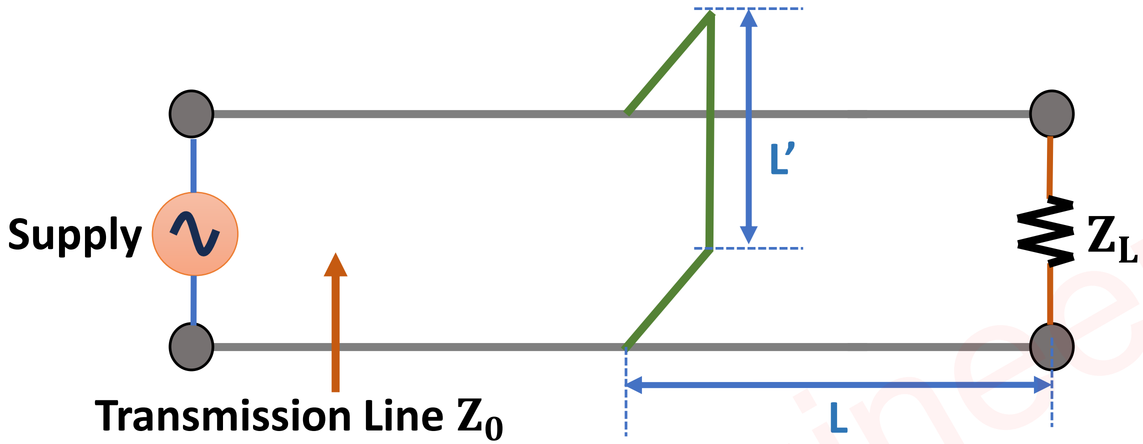
$$\Rightarrow \cot\beta l' = \frac{Z_L - Z_0}{\sqrt{Z_L Z_0}}$$

$$\Rightarrow \tan\beta l' = \frac{\sqrt{Z_L Z_0}}{Z_L - Z_0}$$

$$\Rightarrow \boxed{l' = \frac{\lambda}{2\pi}\tan^{-1}\left(\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0}\right)}$$

Transmission Line Example on Impedance Matching using Single Stub

Question 1 – 100Ω line with air as a dielectric is terminated by load impedance 600Ω and is excited by 1GHz, with match generation. Find the position and length of the single stub.



$\rightarrow Z_0 = 100\Omega$

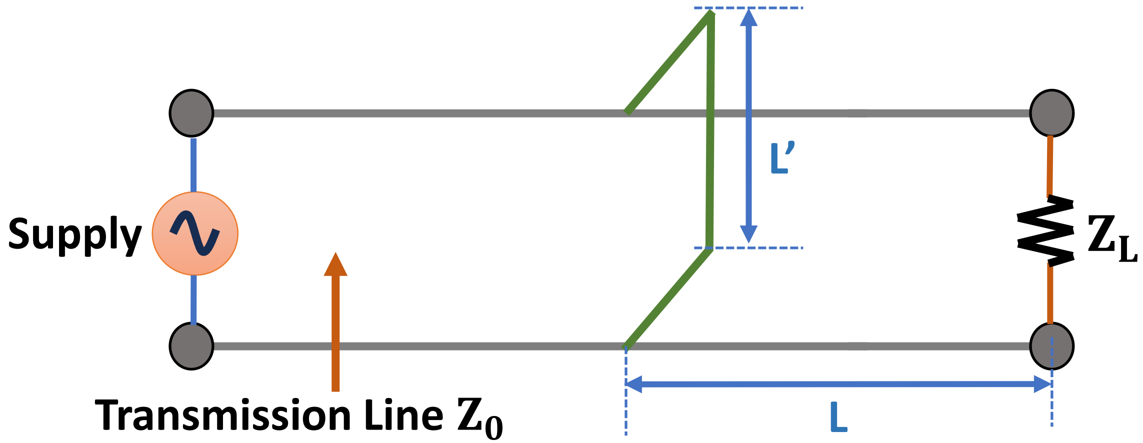
$Z_L = 600\Omega$

$\lambda = \frac{c}{f} = \frac{3\times10^8}{10^9} = 0.3\,m$

$\rightarrow L = \frac{\lambda}{2\pi}\tan^{-1}\sqrt{\frac{Z_L}{Z_0}} = \frac{0.3}{2\pi}\tan^{-1}\sqrt{\frac{600}{100}} = 0.056\,m$

$\rightarrow L' = \frac{\lambda}{2\pi}\tan^{-1}\left(\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0}\right) = \frac{0.3}{2\pi}\tan^{-1}\left(\frac{\sqrt{600\times100}}{500}\right) = 0.0217\,m$

Question 2 – A load (50 + j100)Ω is connected across a 50Ω. Design a short-circuited stub to provide impedance matching at 30MHz.



$\rightsquigarrow$ $Z_L = 50 + j100\ \Omega$

$Z_0 = 50\Omega$

$\lambda = \frac{c}{f} = \frac{3\times10^8}{30\times10^6} = 10m$

$\rightsquigarrow$ $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$

$= \frac{50 + j100 - 50}{50 + j100 + 50}$

$= \frac{j100}{100 + j100} = \frac{100\angle 90}{141.4\angle 45}$

$= 0.707\angle 45$

$\rightsquigarrow$ $|r| = 0.707$, $\phi = 45^\circ$

$\rightsquigarrow$ $L = \dfrac{\lambda}{4\pi}\left[\phi + \pi - \cos^{-1}|r|\right]$

$= \dfrac{10}{4\pi}\left[\dfrac{\pi}{4} + \pi - \cos^{-1} 0.707\right]$

$= \dfrac{10}{4\cancel{\pi}}\left[\cancel{\pi/4} + \cancel{\pi} - \cancel{\pi/4}\right]$

$= 2.5\ m$

$\rightsquigarrow$ $L' = \dfrac{\lambda}{2\pi} \tan^{-1}\left(\dfrac{\sqrt{1+|r|^2}}{2|r|}\right)$

$= \dfrac{10}{2\pi} \tan^{-1}\left(\dfrac{\sqrt{1+|0.707^2|}}{2 \times 0.707}\right)$

$= 1.13\ m$

Transmission Line Example on Losses of Transmission Line

Question 1 – A 50Ω generator feeds a 600Ω Lossless Transmission line. If the line is 200m long and terminated by the load of 500Ω, Determine in dBs

a. Reflection Loss
b. Transmission Loss
c. Return Loss

$$Z_g = 50\Omega$$
$$\alpha = 0$$
$$Z_0 = 600\Omega$$
$$l = 200\,m$$
$$Z_L = 500\Omega$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{500 - 600}{500 + 600} = \frac{-100}{1100} = -0.09$$

$$L_{ref} = 10 \log\left[\frac{1}{1 - |\Gamma^2|}\right] = 10 \log\left[\frac{1}{1 - 0.09^2}\right] = 0.035\ dB$$

$$\rightsquigarrow \quad L_{Trans} = L_{Att} + L_{Ref}$$

$$= 8.686\,\alpha l + 10\log\left[\frac{1}{1-|\Gamma|^2}\right]$$

$$= 0.035 \text{ dB}$$

$$\rightsquigarrow \quad L_{Ret} = 20\log|\Gamma|$$

$$= 20\log 0.09$$

$$= -20.91 \text{ dB}$$

Transmission Line Examples

Question 1 – For a Transmission Line, RLGC Components are given by 5Ω, 1mH, 0.1℧, and 1μF, respectively. Find Z_0, ϒ, α, and β at 1MHz.

$R = 5\Omega$

$L = 1\,mH$

$G = 0.1\,℧$

$C = 1\,\mu F$

$f = 1\,MHz$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$= \sqrt{\frac{5 + j\,2\pi \times 10^6 \times 10^{-3}}{0.1 + j\,2\pi \times 10^6 \times 10^{-6}}}$$

$$= \sqrt{\frac{6.28 \times 10^3 \angle 89.95}{6.28 \angle 89.08}}$$

$$= 31.62 \angle 0.44$$

$$\gamma = \alpha + j\beta$$

$$= \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \sqrt{6.28 \times 10^3 \times 6.28 \times \angle (89.95 + 89.08)}$$

$$= 198 \angle 89.52$$

$$= \underbrace{1.65}_{\alpha} + j\,\underbrace{198}_{\beta}$$

Question 2 – A Lossless Transmission Line of 50Ω is having $V_{MAX} = 2.5V$, $V_{MIN} = 1V$. When terminated by unknown load. Find VSWR, Reflection Coefficient and Load Impedance.

$\rightsquigarrow Z_0 = 50\Omega$

$V_{max} = 2.5V$

$V_{min} = 1V$

$\rightsquigarrow VSWR = \frac{V_{max}}{V_{min}} = \frac{2.5}{1} = 2.5$

$\rightsquigarrow \Gamma = \frac{VSWR-1}{VSWR+1} = \frac{2.5-1}{2.5+1} = \frac{1.5}{3.5} = 0.428$

$\rightsquigarrow VSWR = \frac{Z_L}{Z_0} \quad (Z_L > Z_0)$

$= \frac{Z_0}{Z_L} \quad (Z_0 > Z_L)$

$\Rightarrow 2.5 = \frac{Z_L}{50} \Rightarrow \boxed{Z_L = 125\Omega}$

$\rightsquigarrow \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$

$\Rightarrow 0.428 = \frac{Z_L - 50}{Z_L + 50}$

$\Rightarrow 0.428 Z_L + 50 \times 0.428 = Z_L - 50$

$\Rightarrow Z_L - 0.428 Z_L = 50 \times 0.428 + 50$

$\Rightarrow \boxed{Z_L = 125\Omega}$

Transmission Line Examples

Question 3 – The primary line constants of a Transmission Line per km are specified as R=42.9Ω, L=0.7mH, G=0.1μ℧, and C=0.1μF, respectively. Find Z_0, Υ, α, β and v_p at 5000rad/sec.

$\rightarrow$ $R = 42.9\ \Omega$

$L = 0.7\ mH$

$G = 0.1\ \mu ℧$

$C = 0.1\ \mu F$

$\omega = 5000\ rad/sec$

$\rightarrow$ $Z_0 = \sqrt{\dfrac{R + j\omega L}{G + j\omega C}}$

$= \sqrt{\dfrac{42.9 + j\,5000 \times 0.7 \times 10^{-3}}{10^{-7} + j\,5000 \times 10^{-7}}}$

$= \sqrt{\dfrac{43.04 \angle 4.66}{5 \times 10^{-4} \angle 89.72}}$

$= 293.39 \angle -42.53\ \Omega$

$\rightarrow$ $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$

$= \sqrt{43.04 \times 5 \times 10^{-4} \angle (4.66 + 89.72)}$

$$= 0.14 \angle 47.19$$

$$= \underbrace{0.095}_{\alpha} + j\,\underbrace{0.1027}_{\beta}$$

$$[Np/km] \qquad [Rad/km]$$

$$\Rightarrow v_p = \frac{\omega}{\beta} \quad \frac{[Rad/sec]}{[Rad/km]} = [km/sec]$$

$$= \frac{5000}{0.1027}$$

$$= 48.68 \times 10^{3} \; km/sec$$

Question 4 – A Transmission Line has the following parameters:
$R=2\Omega/m$, $L=8nH/m$, $G=0.5m\mho/m$, $C=0.23pF/m$, and $f=1GHz$
Find Z_0, Υ, α, β and v_p.

$R = 2\ \Omega/m$

$L = 8\ nH/m$

$G = 0.5\ m\mho/m$

$C = 0.23\ pF/m$

$$\rightarrow Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$= \sqrt{\frac{2+j\,2\times 3.14\times 10^9\times 8\times 10^{-9}}{0.5\times 10^{-3}+j\,2\times 3.14\times 10^9\times 0.23\times 10^{-12}}}$$

$$= \sqrt{\frac{50.27\ \angle 87.72}{1.53\times 10^{-3}\angle 70.91}}$$

$$= 181.26\ \angle 8.4\ \Omega$$

$$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$$

$$= \sqrt{50.27\times 1.53\times 10^{-3}\angle(87.72+70.91)}$$

$$= 0.2773 \angle 79.31$$

$$= \underbrace{0.0514}_{\substack{\alpha \\ [Np/m]}} + j\,\underbrace{0.272}_{\substack{\beta \\ [Rad/m]}}$$

$$\rightsquigarrow \quad v_p = \frac{\omega}{\beta} \qquad \frac{[Rad/sec]}{[Rad/m]} = m/sec$$

$$= \frac{2\pi f}{\beta}$$

$$= \frac{2\pi \times 10^{9}}{0.272}$$

$$= 2.308 \times 10^{10}\ m/sec$$